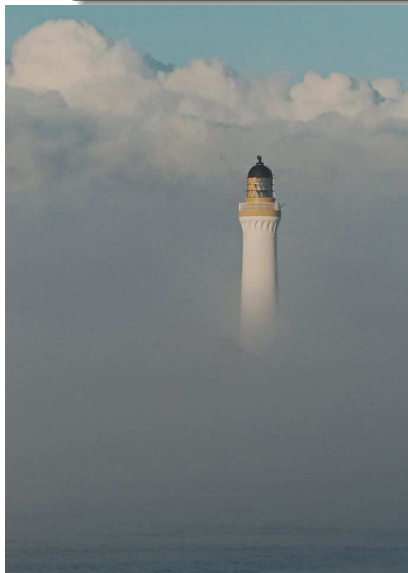
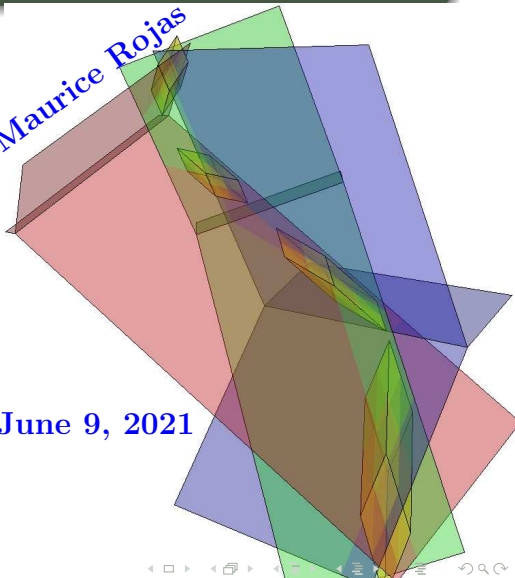


Counting Real Roots in Polynomial-Time for Systems Supported on Circuits



J. Maurice Rojas

June 9, 2021



1 Motivation & Background



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2 Fewnomials and Number Theory



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- 2 Fewnomials and Number Theory
- 3 Faster Real Root Counting for Circuit Systems



We'd like to solve...

...systems like:

$$\begin{aligned} & \left(2x_1^{36}x_2^{194}x_3^{50}x_4^{82}x_5^{60} + x_1^{76}x_2^{240}x_4^{41}x_5 + x_1^{74}x_2^{179}x_3^{25}x_5^{57} + x_1^{25}x_2^{203}x_3^{44}x_4 + x_1^{20}x_2^{167}x_3^{64}x_4^{12}x_5^{68} - 37137x_1^{58}x_2^{194}x_3^{24}x_4^{36}x_5^{25} - \frac{9}{2}x_3^{166}x_4^{68}x_5^{343}, \right. \\ & x_1^{36}x_2^{194}x_3^{50}x_4^{82}x_5^{60} + 2x_1^{76}x_2^{240}x_4^{41}x_5 + x_1^{74}x_2^{179}x_3^{25}x_5^{57} + x_1^{25}x_2^{203}x_3^{44}x_4 + x_1^{20}x_2^{167}x_3^{64}x_4^{12}x_5^{68} - 24849x_1^{58}x_2^{194}x_3^{24}x_4^{36}x_5^{25} - \frac{21}{4}x_3^{166}x_4^{68}x_5^{343}, \\ & x_1^{36}x_2^{194}x_3^{50}x_4^{82}x_5^{60} + x_1^{76}x_2^{240}x_4^{41}x_5 + 2x_1^{74}x_2^{179}x_3^{25}x_5^{57} + x_1^{25}x_2^{203}x_3^{44}x_4 + x_1^{20}x_2^{167}x_3^{64}x_4^{12}x_5^{68} - 21009x_1^{58}x_2^{194}x_3^{24}x_4^{36}x_5^{25} - \frac{21}{4}x_3^{166}x_4^{68}x_5^{343}, \\ & x_1^{36}x_2^{194}x_3^{50}x_4^{82}x_5^{60} + x_1^{76}x_2^{240}x_4^{41}x_5 + x_1^{74}x_2^{179}x_3^{25}x_5^{57} + 2x_1^{25}x_2^{203}x_3^{44}x_4 + x_1^{20}x_2^{167}x_3^{64}x_4^{12}x_5^{68} - 20769x_1^{58}x_2^{194}x_3^{24}x_4^{36}x_5^{25} - \frac{21}{4}x_3^{166}x_4^{68}x_5^{343}, \\ & \left. x_1^{36}x_2^{194}x_3^{50}x_4^{82}x_5^{60} + x_1^{76}x_2^{240}x_4^{41}x_5 + x_1^{74}x_2^{179}x_3^{25}x_5^{57} + x_1^{25}x_2^{203}x_3^{44}x_4 + 2x_1^{20}x_2^{167}x_3^{64}x_4^{12}x_5^{68} - 20754x_1^{58}x_2^{194}x_3^{24}x_4^{36}x_5^{25} - \frac{21}{4}x_3^{166}x_4^{68}x_5^{343} \right) \end{aligned}$$

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But how?

More importantly, why?



Polynomials Equations and Complexity Theory

- Complexity theory is the friend of cryptography,



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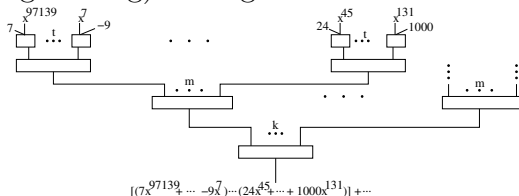
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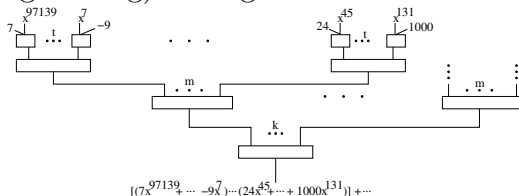
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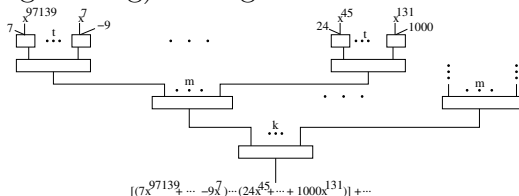


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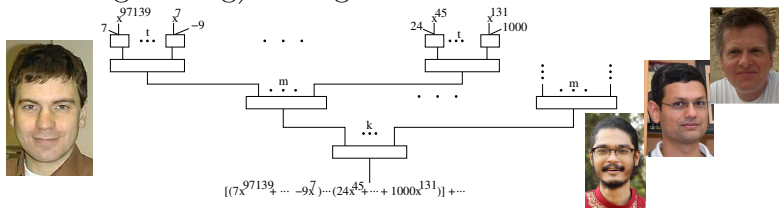


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Fundamental Idea: Theorems about existence of roots are close to the **P** vs. **NP** Problem. Theorems about the deeper structure of polynomials are close to derandomization, i.e., the **P** vs. **BPP** Problem [Koiran, '11; Dutta, Saxena, Thierauf, '20].



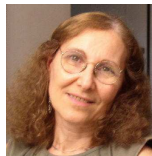
Descartes' Rule, Biochemistry, and Learning Theory...



Recent work on
chemical reaction
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Recent work on chemical reaction networks makes serious use of *exact* counting of real roots

for sparse systems [Bihan, Dickenstein, Giaroli, Shiu, 2019–2020].



Descartes' Rule, Biochemistry, and Learning Theory...



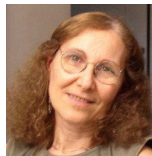
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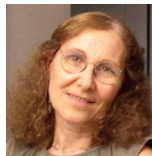
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But what about *exact counting*?



Detecting Real Roots is Already Hard for 1 Sparse (Multivariate) Polynomial!

Complexity Lower Bound.

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An $O(n)$ -time algorithm (assuming affinely independent exponents) is possible!
- But what about systems?



Counting Roots for $(n + 1)$ -nomial $n \times n$ Systems is Easy, but...

Bonus Exercise. *Given $[c_{i,j}] \in \{-H, \dots, H\}^{n \times (n+1)}$,*



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Binomials and (Integer!) Linear Algebra

Solving binomial systems...

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...is the same as **Hermite factorization**:

$$\begin{bmatrix} 1 & 6 & 2 & 6 \\ 7 & 4 & 3 & 4 \\ 7 & 9 & 2 & 8 \\ 4 & 6 & 6 & 5 \end{bmatrix} = AI_4$$



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- *Smith Factorization* helps, and you'll ultimately need (polynomial-time!) *Linear Programming* for the case ∞ !
- $(n + 2)$ -nomial $n \times n$ systems are the next step up, and the last tractable case (for now)...



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Over 245 million complex roots!

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[Bihan, Sottile, 2007] Let $c_{i,j} \in \mathbb{R}$ for all i, j , let $a_1, \dots, a_{n+k} \in \mathbb{R}^n$ be any points *not* all lying in an affine hyperplane, and write $x^{a_i} = x_1^{a_{1,i}} \cdots x_n^{a_{n,i}}$.

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$$F := (f_1, \dots, f_n) = \begin{cases} c_{1,1}x^{a_1} + \cdots + c_{1,n+k}x^{a_{n+k}} \\ \vdots \\ c_{n,1}x^{a_1} + \cdots + c_{n,n+k}x^{a_{n+k}} \end{cases}$$

has no more than $\frac{e^2+3}{4} 2^{(k-1)(k-2)/2} n^{k-1}$ non-degenerate roots in $\mathbb{R}_+^n$.

[Bertrand, Bihan, Sottile '06]: Tight bound for $k=2$ of $n+1$.



Fewnomials and the Quest for Tight Bounds: 2010s



Theorem. [Bürgisser, Ergür, Tonelli-Cueto, 2019] *For an $(n + k)$ -nomial $n \times n$ system with independent standard real Gaussian coefficients*



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Theorem. [Bürgisser, Ergür, Tonelli-Cueto, 2019] *For an $(n + k)$ -nomial $n \times n$ system with independent standard real Gaussian coefficients and fixed support not lying in an affine hyperplane, the average number of roots in $(\mathbb{R}^*)^n$ is $\leq \frac{1}{2} \binom{n+k}{k}$.*



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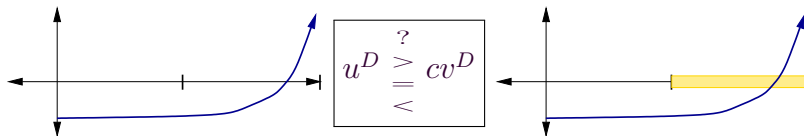
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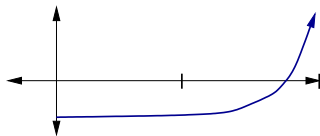
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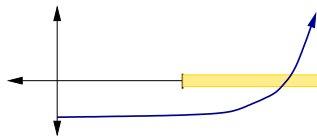


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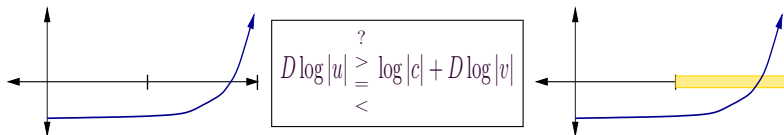


Trouble?



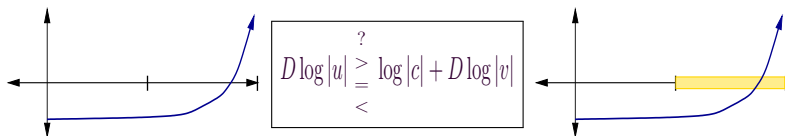
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[Baker, 1966] If $a_i, b_i \in \mathbb{Z}$ with $A := \max_i \log |a_i|$, $B := \max_i \log |b_i|$, and $\Lambda := \sum_{i=1}^n b_i \log a_i$, then $\Lambda \neq 0 \implies |\log |\Lambda|| = O(A)^n \log B$.



Real Univariate Trinomials?

- Deciding the sign of a trinomial $f \in \mathbb{Q}[x_1]$ at $z \in \mathbb{Q}$ in time $(\text{size}(f) + \text{height}(z))^{O(1)}$ is an open problem!



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- Despite nice progress by [Jindal, Sagraloff, 2017] on *coarse* approximants, real root counting in time $(t \log(dH))^{O(1)}$ is still an open problem!



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Main Theorem. [Rojas, 2020] *For any $(n + 2)$ -nomial $n \times n$ system F over $\mathbb{Q}$,*



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- Sufficiently refined versions of the *abc-Conjecture* can reduce the complexity to polynomial in n ...



Key Idea #1: Gaussian Elimination

By Gauss-Jordan Elimination applied to the linear combinations of monomials, our original 5×5 example can be reduced to

$$\begin{aligned}
 x_1^{36} x_2^{194} x_3^{-116} x_4^{14} x_5^{-283} &= 16384 c x_1^{58} x_2^{194} x_3^{-142} x_4^{-32} x_5^{-318} + \frac{1}{4} \\
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Next: Observe that the set of exponent vectors is a *circuit in the sense of combinatorics*: It is the union of a point and the vertex set of a simplex...



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provided some additional sign conditions are met...

Note: The exponents are usually *much* larger!



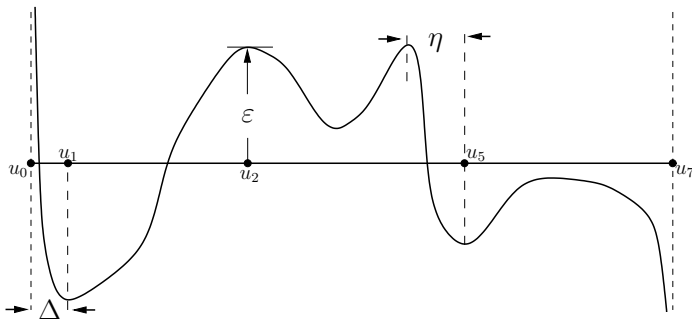
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Examine the graph of the linear combination of logarithms...



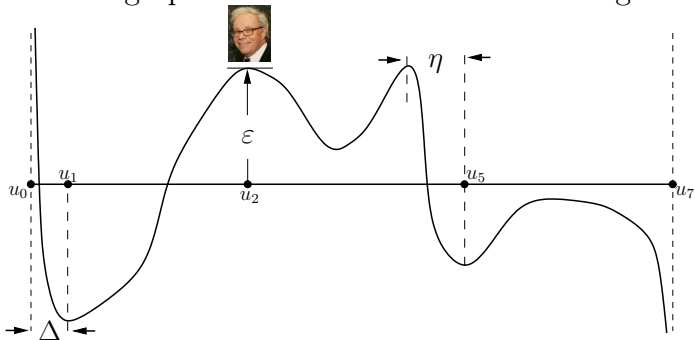
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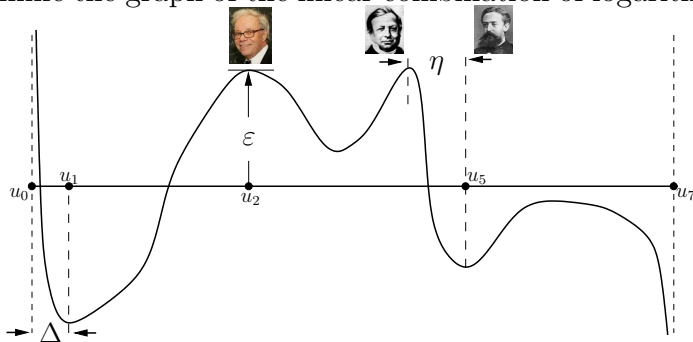


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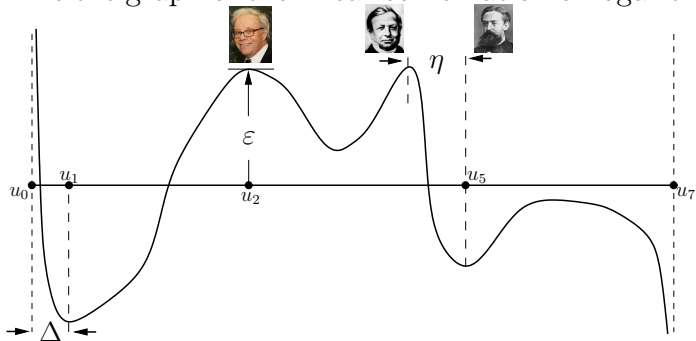


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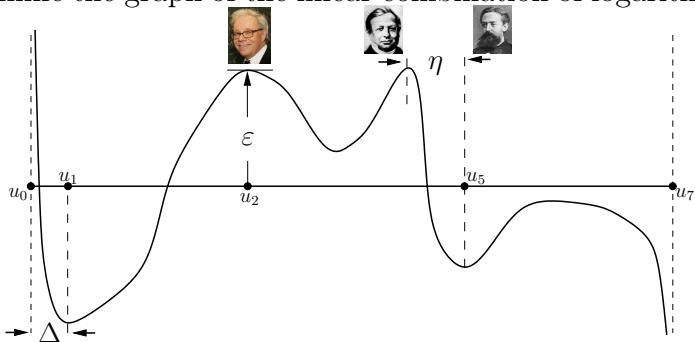


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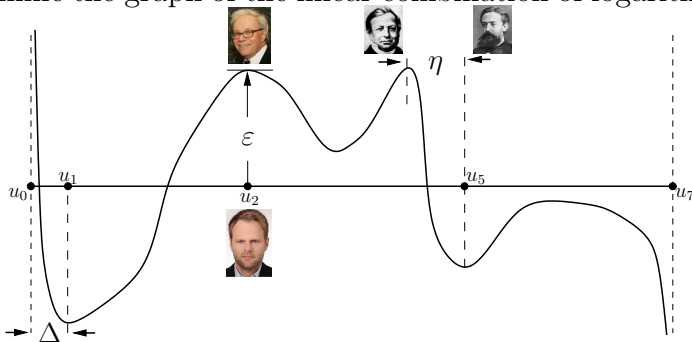


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Upshot: Randomize!

Even though there *are* $(n + 1)$ -nomial $n \times n$ systems,



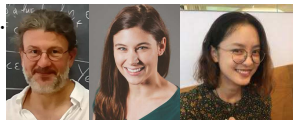
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Even though there *are* $(n + 1)$ -nomial $n \times n$ systems, and *univariate tetranomials* with exponentially close real roots,



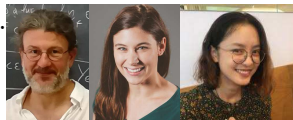
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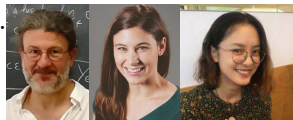


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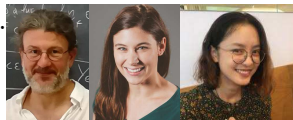


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Thank you for your attention!

See www.math.tamu.edu/~rojas for preprints and further info...

